

Slip-Cast Ceramic Udu Drum Family

(UDU-001) — Build Packet Print Packet

Generated: 2026-05-02

Packet folder: ``/sessions/charming-bold-fermat/mnt/GitHub/udu``

File Map

File	Purpose
<code>`design.md`</code>	Project intent, catalog metadata, assumptions, and validation plan.
<code>`bom.csv`</code>	Starter bill of materials with part categories, quantities, drawing refs, and notes.
<code>`sourcing.csv`</code>	Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
<code>`cut-list.csv`</code>	Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
<code>`drawing-brief.md`</code>	Manufacturing drawing and technical product sketch brief.
<code>`assembly-manual.md`</code>	Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
<code>`validation.csv`</code>	Target/measured values, tolerance, environment, result, and tuning/build action log.
<code>`supplier-rfq.md`</code>	Supplier email/request-for-quote starter.
<code>`visual-bom-brief.md`</code>	Art direction for an image-forward visual BOM.
<code>`wolfram-starter.wl`</code>	Wolfram starter for physics, optimization, visualization, and validation.
<code>`README.md`</code>	Project artifact.
<code>`capstone-deck.md`</code>	Project artifact.
<code>`print-packet.md`</code>	Project artifact.

design.md

Project intent, catalog metadata, assumptions, and validation plan.

Udu Drum Family Build Packet

Source

- Design table: `docs/Udu.xlsx`
- Sheet: `Udu Drum Family`
- Inspected range: `A1:H146`
- Workbook content observed: design inputs, dual Helmholtz calculator, four-size family, variants, playing techniques, ceramic production workflow, BOM, design notes, and Wolfram notebook notes.

Design Intent

Build a slip-cast ceramic udu family with repeatable chamber volumes and two playable Helmholtz tones: the main mouth tone and the side-hole slap tone. The first production target is the medium standard udu, followed by a scaled family.

Governing Model

The udu uses a shared chamber with two main ports:

$$\begin{aligned} f_{\text{top}} &= c/(2\pi) * \text{sqrt}(A_{\text{top}}/(V * L_{\text{top}})) \\ f_{\text{side}} &= c/(2\pi) * \text{sqrt}(A_{\text{side}}/(V * L_{\text{side}})) \end{aligned}$$

When both ports are open and the player's hands move, the modes couple. The single-port formulas are useful first-order targets; measured prototypes should define the final tuning corrections.

Current Workbook Inputs

Field	Current value
Body diameter	10 in

Body height	12 in
Wall thickness	0.3 in
Mouth diameter	3 in
Side hole diameter	2 in
Shape factor	0.7
Clay body	Cone 6 stoneware
Shrinkage	0.12
Speed of sound	13510 in/s

Current Family Targets

Model	Diameter	Height	Volume	Mouth tone	Side tone	Use
Small	8 in	10 in	235 in ³	D#4	A3	Tabletop and high tones
Medium	10 in	12 in	440 in ³	B3	G3	Most versatile standard size
Large	12 in	14 in	739 in ³	G#3	D3	Deep bass and traditional feel
XL	14 in	16 in	1149 in ³	F3	C3	Concert bass, likely needs stand

Critical Design Features

- Body volume controls both main tones; slip casting is valuable because it makes volume repeatable.
- Port diameters and effective neck lengths are the main pitch controls.
- Wall thickness controls shell tap tone, durability, drying risk, and weight more than Helmholtz pitch.
- The side hole edge must be comfortable for repeated hand slaps.
- The mouth rim should be rounded enough for comfort but consistent enough for repeatable hand sealing.
- Feet, stand interface, or flat area should be designed before the mold if the instrument needs stability.

Cultural And Product Note

The udu has Igbo roots and is not just a generic ceramic bass pot. Public-facing copy should be respectful, specific, and verified before publication. For Tony's own build docs, keep the lineage note visible and separate it from product marketing.

Mold Strategy

Recommended first production mold:

- Two-piece plaster body mold split near the widest diameter.
- Separate mouth/neck core or hand-cut mouth after demold if that gives cleaner release.
- Side hole cut at leather-hard stage so tuning can be adjusted by cutter size and file work.
- Smooth mold split where hand contact will happen.
- Add subtle foot pads or a stand-contact feature if the final instrument should sit safely.

Master scale factor:

$$\text{scale} = 1/(1 - \text{shrinkage})$$

For 12 percent shrinkage, use approximately `1.136` before adjusting for measured clay data.

Prototype Ladder

Prototype	Goal	Success criteria
UDU-P0 cup/port tile	Validate clay wall/tap behavior	No cracks; clear ceramic tap; clean port edge
UDU-P1 small body	Validate slip casting and side-hole slap	Measurable Helmholtz tone, comfortable slap edge
UDU-P2 medium body	Hit workbook medium target	Mouth and side tones within +/-50 cents after bisque
UDU-P3 water udu	Explore pitch-bending volume	Smooth tilt-bend without leaking or handling risk
UDU-P4 family molds	Scale S/M/L/XL	Predictable pitch trend across sizes

Workbook Improvement Notes

Recommended next workbook additions:

1. Add measured frequency and cents-error fields for mouth, side hole, and both-open condition.
2. Add cast-wall thickness measured at rim, shoulder, belly, and foot.

3. Add water-fill chamber volume after bisque and glaze fire.
4. Add per-size CAD master dimensions and shrinkage-compensated dimensions.
5. Add a coupled-mode experiment table for both ports open, side port covered, and mouth hand-muted.

Open Assumptions

- Workbook costs are estimates and have not been date-checked.
- The side-hole tuning note in the workbook should be validated empirically; for a simple Helmholtz port, enlarging a port usually raises its open-port resonance, but hand coupling can make perceived playing behavior more complicated.
- Exact mold split and pour/drain design depend on final body shape.

bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

item_id	category	item	qty	spec	make_buy	estimated_cost	source_note	drawing
UDU-BOM-001	Clay	Cone 6 casting slip	5 gal	Commercial premixed stoneware casting slip	Buy	\$35-50	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-002	Mold	#1 pottery plaster	50 lb	Pottery plaster for 2-piece mold	Buy	\$25-35	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-003	Master	3D printed master	2 halves	PLA master scaled for shrinkage	Make	\$5-15	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-004	Mold	Mold soap/release	1 bottle	Plaster-compatible release	Buy	\$8-12	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-005	Cutting	Hole cutter set	1 set	1.5 in to 3 in round cutters	Buy	\$10-20	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-006	Firing	Bisque and glaze firing	2 fires	Cone 06 bisque and Cone 6 glaze	Buy	\$20-50/fire	Workbook estimate not date-checked	UDU-VAL-002
UDU-BOM-007	Finish	Exterior glaze	1 pint	Cone 6 matte or satin glaze	Buy	\$10-20	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-008	Hardware	Rubber feet	1 set	Three low-profile feet	Buy	\$3-5	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-009	Electronics	Contact pickup	1	Piezo disc/contact mic	Buy	\$15-30	Workbook estimate not	UDU-DRW-00

							date- checked	
UDU- BOM-010	Measurement	Water-fill and frequency measurement kit	1	Graduated vessel plus tuner/mic	Buy	TBD	Add supplier/ date before purchase	UDU- VAL-003

sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

item_id	item	required_spec	search_terms	supplier_candidates	date_checked	unit_price	lead_time
UDU-SRC-001	Cone 6 casting slip	Stoneware casting slip with known shrinkage and mature Cone 6 schedule	cone 6 stoneware casting slip 5 gallon shrinkage	TBD			
UDU-SRC-002	#1 pottery plaster	Absorbent pottery mold plaster in 50 lb quantity	USG #1 pottery plaster 50 lb	TBD			
UDU-SRC-003	PLA filament	Stable print material for large master halves	PLA filament large format mold master halves	TBD			
UDU-SRC-004	Mold release	Plaster-compatible parting/release compound	pottery mold soap plaster release	TBD			
UDU-SRC-005	Hole cutters	Clean round cutters from 1.5 in to 3 in	ceramic hole cutter round clay 2 inch 3 inch	TBD			
UDU-SRC-006	Rubber feet	Low profile feet with adhesive or mechanical mount	rubber feet ceramic instrument low profile	TBD			
UDU-SRC-007	Piezo pickup	Contact pickup that can survive installation method	piezo disc contact pickup drum ceramic	TBD			

cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

cut_id	part	qty	rough_dimensions_in	final_dimensions_in	material	orientation	operations
UDU-CUT-001	Master body upper half (3D-printed) — Medium	1	12 x 12 x 7 envelope	master scaled by 1/ (1-shrinkage); ~1.136x at 12% shrinkage	PLA or sealable resin	Z up; mark mouth axis + side-hole datum	Print in session; sand la lines; f voids;
UDU-CUT-002	Master body lower half (3D-printed) — Medium	1	12 x 12 x 7 envelope	mirror of UDU-CUT-001	PLA or sealable resin	Z up; mark split-line datum	Print s seal
UDU-CUT-003	Wooden cottle boards (mold making)	4	18 x 16 x 0.75 each	16 x 14 x 0.75 each	Melamine-faced MDF	Smooth face inward; mark inner clamping zone	Cut ed seal dr clampi holes
UDU-CUT-004	Plaster mother mold half (poured) — Medium	2	16 x 14 x 4 each	16 x 14 x 3.5-3.75 each (around master)	USG #1 pottery plaster (or equiv.)	Pour with master keyed to first half; second half over registration	Mix to 73:100 water; (by we 2-min : pour a sealed master
UDU-CUT-005	Greenware vessel (slip-cast) — Medium	1 per pour	master cavity wet (~12 x 14 x 7 OD)	~10.6 x 12.4 x 6.2 fired (12% shrink)	Cone 6 stoneware casting slip	Mouth axis along master split	Slip po drain a target (~0.30 7.6 mm demol leather
UDU-CUT-006	Mouth opening (cast as part of master)	1	N/A	3.0 in dia FIRED (=3.41 in master)	Cone 6 greenware	Top of body — central	Cast c via mo shape; round leather
UDU-CUT-007		1 per body	N/A	2.0 in dia FIRED	Cone 6 greenware	Side of body at	Brass cutter

	Side hole (leather-hard cut)					hand-comfort height	hole p at leath hard; c burr wi damp
UDU-CUT-008	Foot pads (optional — cast-in or applied)	3 per body	N/A	~0.5 in dia x 0.10 in tall	Rubber adhesive feet OR cast-in clay foot	Tripod 120 deg around base	Apply glaze f OR ca part of master
UDU-CUT-009	Post-bisque tuning trim	as needed	N/A	Within +/-50 cents of target frequency per validation.csv	Bisqued ceramic	N/A	Diamon burr in steps; enlarg mouth side ho raise p
UDU-CUT-010	Wax resist mask (pre-glaze)	1 application	N/A	~0.020 in dry film over mouth rim + side hole edge + interior	Brushable wax resist	N/A	Brush- dry per supplie check openin clear
UDU-CUT-011	Glaze application (exterior only)	1 application	N/A	~0.010-0.020 in per coat; 1 coat exterior matte	Cone 6 satin or matte glaze	Avoid waxed regions and mouth/ side-hole interiors	Dip / b spray exterior
UDU-CUT-012	Family scaling — Small / Large / XL master variants	1 set each	Per design table family-spec rows	Per design table fired dimensions	Same as UDU-CUT-001/002	Per family	Print/m mold p pipelin once Medium voicing stable

drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Udu Drum Family Drawing Brief

Required Views

- Front view with overall height, max body diameter, mouth diameter, and side-hole position.
- Side view showing body profile, wall target, side-hole axis, and foot/stand contact.
- Top view showing mouth opening, rim radius, and symmetry datums.
- Section view through mouth and side hole showing wall thickness and effective neck length.
- Mold split view with registration keys, plaster thickness, pour/drain direction, and seam-cleanup zone.
- Optional electric variant view showing piezo location, wire path, jack location, and service access.

Critical Dimensions

Dimension	Source	Tolerance intent
Chamber volume	Workbook formula and water-fill measurement	Tuning critical
Body diameter and height	Workbook inputs	Tuning and ergonomics
Mouth diameter	Workbook input	Tuning and hand seal
Side-hole diameter	Workbook input and tuning log	Tuning and playability
Wall thickness	Workbook input and cast measurement	Tap tone/durability
Rim radii	CAD and ergonomic test	Comfort
Master scale factor	Measured clay shrinkage	Process critical

Notes For CAD

- Use a named shape factor or profile parameter so S/M/L/XL can be regenerated from the same model.
- Keep side-hole position as a parameter tied to hand comfort, not only aesthetics.

- Design any feet as optional variants so acoustic damping can be tested.
- Keep the water-udu variant separate because water fill and leak-tightness change requirements.

assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Udu Drum Family Assembly Manual

Scope

This manual covers slip-cast ceramic udu prototypes based on `docs/Udu.xlsx`, starting with the medium standard udu and expanding to a scaled family.

Tools

- CAD package for body profile and mold split.
- Large-format 3D printer or outsourced print for master halves.
- Plaster mold-making setup: cottle boards, scale, bucket, plaster, mold soap, clamps.
- Cone 6 stoneware casting slip.
- Hole cutters, ribs, sponges, fettling knife, rasp, and small files.
- Calipers, wall-thickness gauge if available, graduated water-fill vessel, tuner/microphone.
- Kiln access, wax resist, exterior glaze.
- Optional: rubber feet, stand prototype, piezo/contact mic.

Process

1. **Choose size**

- Start with the medium workbook design unless testing mold workflow at smaller scale.
- Assign build ID and expected shrinkage.

2. **CAD body and mold strategy**

- Model the target fired body first.
- Add shrinkage scale to produce the master body.
- Choose a split line that releases cleanly and avoids hand-contact ridges.

3. ****Print and seal master****

- Print master halves rigid enough for plaster.
- Sand high layer lines, fill defects, and seal.
- Verify master diameter and height before molding.

4. ****Make plaster mold****

- Apply release.
- Pour mold halves with registration keys.
- Dry completely before first casting.
- Record mold dry weight and first-use date.

5. ****Slip cast****

- Fill mold with slip.
- Drain after target wall thickness buildup.
- Record fill time, drain time, slip batch, room condition, and demold time.

6. ****Leather-hard work****

- Clean seam.
- Cut side hole undersized or at first target.
- Round mouth and side-hole edges.
- Add foot/stand features only if they do not trap drying stress.

7. ****Dry****

- Dry under plastic for even moisture.
- Rotate periodically if needed.
- Inspect rim, side hole, foot, and seam.

8. ****Bisque fire****

- Fire to the clay body's bisque schedule.
- Measure dimensions, chamber volume, mass, and pitch.

9. ****Tune****

- Tune with small changes and document each pass.
- For simple open-port Helmholtz behavior, larger port area usually raises resonance; hand-covered playing conditions should be measured separately.
- Do not remove more material until the measurement setup is repeatable.

10. ****Glaze****

- Glaze exterior only unless intentionally testing interior glaze damping.
- Keep mouth and side-hole edges clean.

11. ****Final validation****

- Measure final tones, shell tap, weight, stability, and ergonomics.
- Record whether feet or stands damp the ceramic tap tone.

Failure Modes To Watch

- Cracks at side hole: hole cut too late, edge too sharp, or drying too fast.
- Dead bass response: chamber too small, ports poorly proportioned, walls too thick, or hand seal poor.
- Uncomfortable slap edge: rim needs larger radius or smoother fired edge.
- Warped base: unsupported drying or uneven mold thickness.
- Over-damped shell: glaze too heavy, feet too soft, or wall too thick.

validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

build_id	stage	date	model	clay_body	shrinkage_expected_pct	master_scale_factor	body_diam_in
UDU-P1	greenware		Small				8
UDU-P1	bisque		Small				TBD
UDU-P1	bisque		Small				TBD
UDU-P2	greenware		Medium				10
UDU-P2	bisque		Medium				TBD
UDU-P2	bisque		Medium				TBD
UDU-P2	glaze_fire		Medium				TBD
UDU-P3	bisque		Water udu				TBD

supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ — Slip-Cast Ceramic Udu Drum Family (UDU-001 family)

> Use this as a template for outreach to ceramic-supply, plaster, mold-making, and finishing suppliers. Customize the Subject and address fields per recipient.

Subject: RFQ — slip-cast ceramic udu drum prototype materials and consumables

Hello,

I'm a mechanical R&D engineer prototyping a small family of slip-cast ceramic udu drums (Igbo-rooted clay vessel drum, S/M/L/XL sizes, ~8-14 in body diameter, ~0.30 in / 7.6 mm wall) and need quotes for the following. The first build is a Medium standard udu (10 in dia × 12 in tall, target mouth tone B3 ≈ 247 Hz, side-hole tone G3 ≈ 196 Hz) based on the parametric dual-Helmholtz design table at [tonykoop/udu](https://github.com/tonykoop/udu).

Items

#	Item	Required spec	Approx. qty (per prototype run)
--- --- --- ---			
1	Cone 6 stoneware casting slip	Mature Cone 6 stoneware slip with published shrinkage (X/Y/Z), water absorption % , and firing schedule . Larger pour volumes than typical small-vessel work — slip needs to flow predictably into a 12+ in master cavity without cracking on drain.	5 gal (Medium) — 15 gal (S/M/L/XL set)
2	#1 pottery plaster (large mold scale)	USG #1 Pottery Plaster (or equivalent) suitable for a 16 × 14 × 4 in two-piece pour. Need water:plaster ratio , set time , and dimensional stability (creep) over months of use.	50 lb (single Medium mold) — 200 lb (full S/M/L/XL set)
3	Mold release / mold soap	Plaster-compatible parting agent. Larger surface area than small vessels — needs to release cleanly off a 12+ in body without locking.	2 bottles / 32 oz
4			

	Cone 6 exterior glaze	Satin or matte glaze. **Color: TBD** — earth tones / iron oxide / rust red are visually closer to traditional fired Igbo-clay ware. Glaze must NOT bridge mouth or side-hole edges (acoustic-critical).	1-2 pints per body
5	Brushable wax resist	Standard ceramic wax resist for masking mouth rim, side-hole rim, and chamber interior pre-glaze.	1 jar / 16 oz
6	Hole cutter set	Round cutters, **1.5 in to 3 in diameter** , for greenware side-hole cutting. Brass tube cutters or commercial hole-cutter discs both acceptable.	1 set
7	Diamond needle file / burr set (post-bisque tuning)	Assorted profiles, **3-10 mm** ; resin-bonded acceptable. Used to enlarge ports for pitch correction.	1 set
8	Bisque + glaze firing service (if local)	Cone 06 bisque + Cone 6 glaze fire; medium-large kiln (10 ft ³ +) can fit XL family.	2 fires per body
9	(Optional) Master printing / outsource	Large-format PLA or resin print, up to **~14 in master width** with **0.010 in dimensional accuracy** . We currently print in-house but want a backup option.	1-4 master pairs (S/M/L/XL)
10	(Optional) Rubber feet — low-profile	Self-adhesive low-profile rubber feet, ~0.5 in dia, for floor stability. Test-fit only — feet damp shell tap tone, so this is an acoustic trade-off.	3 per body
11	(Optional) Piezo / contact pickup	Contact pickup that can survive being epoxied to interior wall OR magnetically/clip-mounted externally.	1 per electric variant

What we need in your quote

- ****Unit price**** and any volume breaks
- ****Minimum order quantity****
- ****Current stock status**** and ****lead time****
- ****Shipping estimate**** to 94566 (Pleasanton, CA) — or local pickup option
- ****Material safety data sheet (MSDS)**** and ****technical data sheet**** for items 1, 2, 4, 5
- ****Recommended substitutions**** if anything goes out of stock
- For item 1: most recent ****shrinkage test data**** for your batch (we re-measure on our side, but a baseline matters)
- For item 9: ****photos of similar-sized prints**** showing layer-line consistency and dimensional check process

Acceptance criteria

The prototypes are measured for X/Y/Z shrinkage, fired chamber volume (water-fill), wall-thickness consistency at rim/shoulder/belly/foot, and acoustic

tuning at TWO conditions: mouth-tone (mouth open, side covered) and side-hole tone (side open, mouth covered or hand-muted). Target tuning tolerance: ****±50 cents post-bisque**** on both modes for the Medium body. Slip with unknown or wildly variable shrinkage invalidates the master scale factor (currently $\frac{1}{1-0.12} = 1.136$), so reproducibility data matters more than headline price.

The repo is a private R&D portfolio repo; we do not need any IP-restricted documents. Public-domain MSDS and supplier spec sheets are sufficient.

Cultural note for sales / marketing teams

The udu has Igbo (Nigerian) roots and is not a generic ceramic bass pot. We're not asking for marketing positioning — we're asking for materials. But please don't include "decorative African pot" or similar reductive copy in any spec sheet response. Lineage attribution belongs to the player and the documentation, not the supplier.

Thank you,

Tony Koop

Mechanical R&D Engineer

tonykoop@gmail.com

github.com/tonykoop/udu (private; access on request)

visual-bom-brief.md

Art direction for an image-forward visual BOM.

Udu Visual BOM Brief

Goal

Create a one-page visual BOM for the slip-cast udu family that shows the body, mold, ports, finishing choices, and optional electric/water variants.

Layout

- Header: "Slip-Cast Ceramic Udu Drum Family".
- Hero image: medium udu render or finished prototype.
- Family strip: small, medium, large, XL silhouettes with target pitch ranges.
- Exploded/process view: 3D printed master halves, plaster mold halves, cast body, side-hole cut, fired body.
- BOM table: item number, material/tool, quantity, spec, cost estimate, make/buy, image.
- Physics inset: top-port and side-port Helmholtz formulas.
- Measurement inset: body volume, mouth tone, side tone, wall thickness, mass.

Image Requirements

Use real photos as soon as available:

- Printed master.
- Plaster mold halves.
- Leather-hard cast before side-hole cutting.
- Side-hole close-up.
- Fired body with hand scale.
- Optional pickup or rubber-foot detail.

Generated or CAD images should be marked as placeholders until replaced by shop photos.

wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

```
(* Udu drum family Helmholtz notebook starter *)

ClearAll["Global`*"];

c = 13510; (* in/s at about 68 F *)

areaCircle[d_] := Pi*(d/2)^2;
leff[wall_, area_] := wall + 0.6*Sqrt[area/Pi];
volumeOvoid[diam_, height_, shapeFactor_] := Pi/
6*diam^2*height*shapeFactor;
helmholtzHz[area_, volume_, neckLength_] :=
  (c/(2*Pi))*Sqrt[area/(volume*neckLength)];
centsError[measured_, target_] := 1200*Log[2,
measured/target];

uduModel[name_, diam_, height_, wall_, mouthDiam_,
sideDiam_, shapeFactor_] :=
Module[{v, aMouth, aSide, lMouth, lSide},
  v = volumeOvoid[diam, height, shapeFactor];
  aMouth = areaCircle[mouthDiam];
  aSide = areaCircle[sideDiam];
  lMouth = leff[wall, aMouth];
  lSide = leff[wall, aSide];
<|
  "Name" -> name,
  "VolumeIn3" -> v,
  "MouthHz" -> helmholtzHz[aMouth, v, lMouth],
  "SideHz" -> helmholtzHz[aSide, v, lSide],
  "IntervalSemitones" -> 12*Log[2,
helmholtzHz[aMouth, v, lMouth]/helmholtzHz[aSide, v,
lSide]]
|>
];

models = {
  uduModel["Small", 8, 10, 0.3, 3, 2, 0.7],
  uduModel["Medium", 10, 12, 0.3, 3, 2, 0.7],
  uduModel["Large", 12, 14, 0.3, 3, 2, 0.7],
```

```
uduModel["XL", 14, 16, 0.3, 3, 2, 0.7]
};

Dataset[models]

(* First coupled-mode placeholder: replace kCouple
with measured fit. *)
coupledFrequencies[f1_, f2_, kCouple_] :=
  Sqrt[Eigenvalues[{{f1^2, kCouple}, {kCouple,
f2^2}}]]];
```

README.md

Project artifact.

Udu — Slip-Cast Ceramic Vessel Drum Family

> *Engineering documentation for a 3D-printed-master, plaster-mold, cone-6-stoneware slip-cast udu drum family — from dual-Helmholtz physics through the parametric design table to a manufacturable build packet.*

![[Hero — AI-generated concept render of a Medium slip-cast udu (placeholder until first prototype is photographed)]](images/hero-concept.png)

*AI-generated concept render — replace with p

capstone-deck.md

Project artifact.

Slip-Cast Ceramic Udu Drum Family (UDU-001) — Build Packet

- Musical instrument documentation capstone
- Packet: /sessions/charming-bold-fermat/mnt/GitHub/udu
- Generated: 2026-05-02

How To Use This Packet

- Start with design.md for intent and assumptions.
- Use bom.csv, sourcing.csv, and cut-list.csv before buying or cutting.
- Use drawing-brief.md and CAD/CNC folders before machining.
- Print the packet for shopping, shop work, and validation.

File Map

- design.md: Project intent, catalog metadata, assumptions, and validation plan.
- bom.csv: Starter bill of materials with part categories, quantities, drawing refs, and notes.
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- cut-list.csv: Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
- drawing-brief.md: Manufacturing drawing and technical product sketch brief.
- assembly-manual.md: Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
- validation.csv: Target/measured values, tolerance, environment, result, and tuning/build action log.
- supplier-rfq.md: Supplier email/request-for-quote starter.

Build Workflow

- Design and assumptions
- Source materials and hardware
- Prepare stock, fixtures, and CNC/laser/lathe setup
- Assemble, tune, finish, and validate

Sourcing And BOM

- BOM gives part categories and drawing references.
- Sourcing tracks search terms, supplier candidates, price/date, lead time, and substitutions.
- Visual BOM brief turns the parts list into a presentation-ready image board.

Shop Packet

- Cut list for lumber/sheet/blank planning.
- Assembly manual for away-from-keyboard work.
- Validation sheet for measured dimensions, tuning, and pass/fail checks.

Drawings, CAD, CNC

- drawing-brief.md defines required views, dimensions, datums, and sketch intent.
- cad/ holds models and design tables.
- cnc/ holds CAM, toolpaths, setup sheets, and dry-run notes.
- drawings/ holds PDFs, SVGs, DXFs, and drawing exports.

Images And Screenshots

- images/family-overview-concept.png
- images/hero-concept.png

Next Actions

- Replace TBDs with measured/source-backed values.
- Verify live supplier price and availability before buying.
- Export final drawings and visual BOM images.
- Regenerate this deck and print packet after final edits.

print-packet.md

Project artifact.

Slip-Cast Ceramic Udu Drum Family (UDU-001) — Build Packet Print Packet

Generated: 2026-05-02

Packet folder: ``/sessions/charming-bold-fermat/mnt/GitHub/udu``

File Map

File	Purpose
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<code>`bom.csv`</code>	Starter bill of materials with part categories, quantities, drawing refs, and notes.
<code>`sourcing.csv`</code>	Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
<code>`cut-list.csv`</code>	Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
<code>`drawing-brief.md`</code>	Manufacturing drawing and technical product sketch brief.
<code>`assembly-manual.md`</code>	Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
<code>`validation.csv`</code>	Target/measured values, tolerance, environment, result, and tuning/build action log.
<code>`supplier-rfq.md`</code>	Supplier email/request-for-quote starter.
<code>`visual-bom-brief.md`</code>	Art direction for an image-forward visual BOM.
<code>`wolfram-starter.wl`</code>	Wolfram starter for physics, optimization, visualization, and validation.
<code>`README.md`</code>	Project artifact.

design.md

Project intent, catalog metadata, assumptions, and validation plan.

Udu Drum Family Build Packet

Source

- Design table: `docs/Udu.xlsx`
- Sheet: `Udu Drum Family`
- Inspected range: `A1:H146`
- Workbook content observed: design inputs, dual Helmholtz calculator, four-size family, variants, playing techniques, ceramic production workflow, BOM, design notes, and Wolfram notebook notes.

Design Intent

Build a slip-cast ceramic udu family with repeatable chamber volumes and two playable Helmholtz tones: the main mouth tone and the side-hole slap tone. The first production target is the medium standard udu, followed by a scaled family.

Governing Model

The udu uses a shared chamber with two main ports:

$$\begin{aligned} f_{\text{top}} &= c/(2\pi) * \text{sqrt}(A_{\text{top}}/(V * L_{\text{top}})) \\ f_{\text{side}} &= c/(2\pi) * \text{sqrt}(A_{\text{side}}/(V * L_{\text{side}})) \end{aligned}$$

When both ports are open and the player's hands move, the modes couple. The single-port formulas are useful first-order targets; measured prototypes should define the final tuning corrections.

Current Workbook Inputs

Field	Current value
Body diameter	10 in

Body height	12 in
Wall thickness	0.3 in
Mouth diameter	3 in
Side hole diameter	2 in
Shape factor	0.7
Clay body	Cone 6 stoneware
Shrinkage	0.12
Speed of sound	13510 in/s

Current Family Targets

Model	Diameter	Height	Volume	Mouth tone	Side tone	Use
Small	8 in	10 in	235 in ³	D#4	A3	Tabletop and high tones
Medium	10 in	12 in	440 in ³	B3	G3	Most versatile standard size
Large	12 in	14 in	739 in ³	G#3	D3	Deep bass and traditional feel
XL	14 in	16 in	1149 in ³	F3	C3	Concert bass, likely needs stand

Critical Design Features

- Body volume controls both main tones; slip casting is valuable because it makes volume repeatable.
- Port diameters and effective neck lengths are the main pitch controls.
- Wall thickness controls shell tap tone, durability, drying risk, and weight more than Helmholtz pitch.
- The side hole edge must be comfortable for repeated hand slaps.
- The mouth rim should be rounded enough for comfort but consistent enough for repeatable hand sealing.
- Feet, stand interface, or flat area should be designed before the mold if the instrument needs stability.

Cultural And Product Note

The udu has Igbo roots and is not just a generic ceramic bass pot. Public-facing copy should be respectful, specific, and verified before publication. For Tony's own build docs, keep the lineage note visible and separate it from product marketing.

Mold Strategy

Recommended first production mold:

- Two-piece plaster body mold split near the widest diameter.
- Separate mouth/neck core or hand-cut mouth after demold if that gives cleaner release.
- Side hole cut at leather-hard stage so tuning can be adjusted by cutter size and file work.
- Smooth mold split where hand contact will happen.
- Add subtle foot pads or a stand-contact feature if the final instrument should sit safely.

Master scale factor:

$$\text{scale} = 1/(1 - \text{shrinkage})$$

For 12 percent shrinkage, use approximately `1.136` before adjusting for measured clay data.

Prototype Ladder

Prototype	Goal	Success criteria
UDU-P0 cup/port tile	Validate clay wall/tap behavior	No cracks; clear ceramic tap; clean port edge
UDU-P1 small body	Validate slip casting and side-hole slap	Measurable Helmholtz tone, comfortable slap edge
UDU-P2 medium body	Hit workbook medium target	Mouth and side tones within +/-50 cents after bisque
UDU-P3 water udu	Explore pitch-bending volume	Smooth tilt-bend without leaking or handling risk
UDU-P4 family molds	Scale S/M/L/XL	Predictable pitch trend across sizes

Workbook Improvement Notes

Recommended next workbook additions:

1. Add measured frequency and cents-error fields for mouth, side hole, and both-open condition.
2. Add cast-wall thickness measured at rim, shoulder, belly, and foot.

3. Add water-fill chamber volume after bisque and glaze fire.
4. Add per-size CAD master dimensions and shrinkage-compensated dimensions.
5. Add a coupled-mode experiment table for both ports open, side port covered, and mouth hand-muted.

Open Assumptions

- Workbook costs are estimates and have not been date-checked.
- The side-hole tuning note in the workbook should be validated empirically; for a simple Helmholtz port, enlarging a port usually raises its open-port resonance, but hand coupling can make perceived playing behavior more complicated.
- Exact mold split and pour/drain design depend on final body shape.

bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

item_id	category	item	qty	spec	make_buy	estimated_cost	source_note	drawing
UDU-BOM-001	Clay	Cone 6 casting slip	5 gal	Commercial premixed stoneware casting slip	Buy	\$35-50	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-002	Mold	#1 pottery plaster	50 lb	Pottery plaster for 2-piece mold	Buy	\$25-35	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-003	Master	3D printed master	2 halves	PLA master scaled for shrinkage	Make	\$5-15	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-004	Mold	Mold soap/release	1 bottle	Plaster-compatible release	Buy	\$8-12	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-005	Cutting	Hole cutter set	1 set	1.5 in to 3 in round cutters	Buy	\$10-20	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-006	Firing	Bisque and glaze firing	2 fires	Cone 06 bisque and Cone 6 glaze	Buy	\$20-50/fire	Workbook estimate not date-checked	UDU-VAL-002
UDU-BOM-007	Finish	Exterior glaze	1 pint	Cone 6 matte or satin glaze	Buy	\$10-20	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-008	Hardware	Rubber feet	1 set	Three low-profile feet	Buy	\$3-5	Workbook estimate not date-checked	UDU-DRW-00
UDU-BOM-009	Electronics	Contact pickup	1	Piezo disc/contact mic	Buy	\$15-30	Workbook estimate not	UDU-DRW-00

							date- checked	
UDU- BOM-010	Measurement	Water-fill and frequency measurement kit	1	Graduated vessel plus tuner/mic	Buy	TBD	Add supplier/ date before purchase	UDU- VAL-003

sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

item_id	item	required_spec	search_terms	supplier_candidates	date_checked	unit_price	lead_time
UDU-SRC-001	Cone 6 casting slip	Stoneware casting slip with known shrinkage and mature Cone 6 schedule	cone 6 stoneware casting slip 5 gallon shrinkage	TBD			
UDU-SRC-002	#1 pottery plaster	Absorbent pottery mold plaster in 50 lb quantity	USG #1 pottery plaster 50 lb	TBD			
UDU-SRC-003	PLA filament	Stable print material for large master halves	PLA filament large format mold master halves	TBD			
UDU-SRC-004	Mold release	Plaster-compatible parting/release compound	pottery mold soap plaster release	TBD			
UDU-SRC-005	Hole cutters	Clean round cutters from 1.5 in to 3 in	ceramic hole cutter round clay 2 inch 3 inch	TBD			
UDU-SRC-006	Rubber feet	Low profile feet with adhesive or mechanical mount	rubber feet ceramic instrument low profile	TBD			
UDU-SRC-007	Piezo pickup	Contact pickup that can survive installation method	piezo disc contact pickup drum ceramic	TBD			

cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

cut_id	part	qty	rough_dimensions_in	final_dimensions_in	material	orientation	operations
UDU-CUT-001	Master body upper half (3D-printed) — Medium	1	12 x 12 x 7 envelope	master scaled by 1/ (1-shrinkage); ~1.136x at 12% shrinkage	PLA or sealable resin	Z up; mark mouth axis + side-hole datum	Print in session; sand la lines; f voids;
UDU-CUT-002	Master body lower half (3D-printed) — Medium	1	12 x 12 x 7 envelope	mirror of UDU-CUT-001	PLA or sealable resin	Z up; mark split-line datum	Print s seal
UDU-CUT-003	Wooden cottle boards (mold making)	4	18 x 16 x 0.75 each	16 x 14 x 0.75 each	Melamine-faced MDF	Smooth face inward; mark inner clamping zone	Cut ed seal dr clampi holes
UDU-CUT-004	Plaster mother mold half (poured) — Medium	2	16 x 14 x 4 each	16 x 14 x 3.5-3.75 each (around master)	USG #1 pottery plaster (or equiv.)	Pour with master keyed to first half; second half over registration	Mix to 73:100 water; (by we 2-min : pour a sealed master
UDU-CUT-005	Greenware vessel (slip-cast) — Medium	1 per pour	master cavity wet (~12 x 14 x 7 OD)	~10.6 x 12.4 x 6.2 fired (12% shrink)	Cone 6 stoneware casting slip	Mouth axis along master split	Slip po drain a target (~0.30 7.6 mm demol leather
UDU-CUT-006	Mouth opening (cast as part of master)	1	N/A	3.0 in dia FIRED (=3.41 in master)	Cone 6 greenware	Top of body — central	Cast c via mo shape; round leather
UDU-CUT-007		1 per body	N/A	2.0 in dia FIRED	Cone 6 greenware	Side of body at	Brass cutter

	Side hole (leather-hard cut)					hand-comfort height	hole p at leath hard; c burr wi damp
UDU-CUT-008	Foot pads (optional — cast-in or applied)	3 per body	N/A	~0.5 in dia x 0.10 in tall	Rubber adhesive feet OR cast-in clay foot	Tripod 120 deg around base	Apply glaze f OR ca part of master
UDU-CUT-009	Post-bisque tuning trim	as needed	N/A	Within +/-50 cents of target frequency per validation.csv	Bisqued ceramic	N/A	Diamo burr in steps; enlarge mouth side ho raise p
UDU-CUT-010	Wax resist mask (pre-glaze)	1 application	N/A	~0.020 in dry film over mouth rim + side hole edge + interior	Brushable wax resist	N/A	Brush- dry per supplie check openin clear
UDU-CUT-011	Glaze application (exterior only)	1 application	N/A	~0.010-0.020 in per coat; 1 coat exterior matte	Cone 6 satin or matte glaze	Avoid waxed regions and mouth/ side-hole interiors	Dip / b spray exterior
UDU-CUT-012	Family scaling — Small / Large / XL master variants	1 set each	Per design table family-spec rows	Per design table fired dimensions	Same as UDU-CUT-001/002	Per family	Print/m mold p pipelin once Medium voicing stable

drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Udu Drum Family Drawing Brief

Required Views

- Front view with overall height, max body diameter, mouth diameter, and side-hole position.
- Side view showing body profile, wall target, side-hole axis, and foot/stand contact.
- Top view showing mouth opening, rim radius, and symmetry datums.
- Section view through mouth and side hole showing wall thickness and effective neck length.
- Mold split view with registration keys, plaster thickness, pour/drain direction, and seam-cleanup zone.
- Optional electric variant view showing piezo location, wire path, jack location, and service access.

Critical Dimensions

Dimension	Source	Tolerance intent
Chamber volume	Workbook formula and water-fill measurement	Tuning critical
Body diameter and height	Workbook inputs	Tuning and ergonomics
Mouth diameter	Workbook input	Tuning and hand seal
Side-hole diameter	Workbook input and tuning log	Tuning and playability
Wall thickness	Workbook input and cast measurement	Tap tone/durability
Rim radii	CAD and ergonomic test	Comfort
Master scale factor	Measured clay shrinkage	Process critical

Notes For CAD

- Use a named shape factor or profile parameter so S/M/L/XL can be regenerated from the same model.
- Keep side-hole position as a parameter tied to hand comfort, not only aesthetics.

- Design any feet as optional variants so acoustic damping can be tested.
- Keep the water-udu variant separate because water fill and leak-tightness change requirements.

assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Udu Drum Family Assembly Manual

Scope

This manual covers slip-cast ceramic udu prototypes based on `docs/Udu.xlsx`, starting with the medium standard udu and expanding to a scaled family.

Tools

- CAD package for body profile and mold split.
- Large-format 3D printer or outsourced print for master halves.
- Plaster mold-making setup: cottle boards, scale, bucket, plaster, mold soap, clamps.
- Cone 6 stoneware casting slip.
- Hole cutters, ribs, sponges, fettling knife, rasp, and small files.
- Calipers, wall-thickness gauge if available, graduated water-fill vessel, tuner/microphone.
- Kiln access, wax resist, exterior glaze.
- Optional: rubber feet, stand prototype, piezo/contact mic.

Process

1. **Choose size**

- Start with the medium workbook design unless testing mold workflow at smaller scale.
- Assign build ID and expected shrinkage.

2. **CAD body and mold strategy**

- Model the target fired body first.
- Add shrinkage scale to produce the master body.
- Choose a split line that releases cleanly and avoids hand-contact ridges.

3. ****Print and seal master****

- Print master halves rigid enough for plaster.
- Sand high layer lines, fill defects, and seal.
- Verify master diameter and height before molding.

4. ****Make plaster mold****

- Apply release.
- Pour mold halves with registration keys.
- Dry completely before first casting.
- Record mold dry weight and first-use date.

5. ****Slip cast****

- Fill mold with slip.
- Drain after target wall thickness buildup.
- Record fill time, drain time, slip batch, room condition, and demold time.

6. ****Leather-hard work****

- Clean seam.
- Cut side hole undersized or at first target.
- Round mouth and side-hole edges.
- Add foot/stand features only if they do not trap drying stress.

7. ****Dry****

- Dry under plastic for even moisture.
- Rotate periodically if needed.
- Inspect rim, side hole, foot, and seam.

8. ****Bisque fire****

- Fire to the clay body's bisque schedule.
- Measure dimensions, chamber volume, mass, and pitch.

9. ****Tune****

- Tune with small changes and document each pass.
- For simple open-port Helmholtz behavior, larger port area usually raises resonance; hand-covered playing conditions should be measured separately.
- Do not remove more material until the measurement setup is repeatable.

10. ****Glaze****

- Glaze exterior only unless intentionally testing interior glaze damping.
- Keep mouth and side-hole edges clean.

11. ****Final validation****

- Measure final tones, shell tap, weight, stability, and ergonomics.
- Record whether feet or stands damp the ceramic tap tone.

Failure Modes To Watch

- Cracks at side hole: hole cut too late, edge too sharp, or drying too fast.
- Dead bass response: chamber too small, ports poorly proportioned, walls too thick, or hand seal poor.
- Uncomfortable slap edge: rim needs larger radius or smoother fired edge.
- Warped base: unsupported drying or uneven mold thickness.
- Over-damped shell: glaze too heavy, feet too soft, or wall too thick.

validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

build_id	stage	date	model	clay_body	shrinkage_expected_pct	master_scale_factor	body_diam_in
UDU-P1	greenware		Small				8
UDU-P1	bisque		Small				TBD
UDU-P1	bisque		Small				TBD
UDU-P2	greenware		Medium				10
UDU-P2	bisque		Medium				TBD
UDU-P2	bisque		Medium				TBD
UDU-P2	glaze_fire		Medium				TBD
UDU-P3	bisque		Water udu				TBD

supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ — Slip-Cast Ceramic Udu Drum Family (UDU-001 family)

> Use this as a template for outreach to ceramic-supply, plaster, mold-making, and finishing suppliers. Customize the Subject and address fields per recipient.

Subject: RFQ — slip-cast ceramic udu drum prototype materials and consumables

Hello,

I'm a mechanical R&D engineer prototyping a small family of slip-cast ceramic udu drums (Igbo-rooted clay vessel drum, S/M/L/XL sizes, ~8-14 in body diameter, ~0.30 in / 7.6 mm wall) and need quotes for the following. The first build is a Medium standard udu (10 in dia × 12 in tall, target mouth tone B3 ≈ 247 Hz, side-hole tone G3 ≈ 196 Hz) based on the parametric dual-Helmholtz design table at [tonykoop/udu](https://github.com/tonykoop/udu).

Items

#	Item	Required spec	Approx. qty (per prototype run)
--- --- --- ---			
1	Cone 6 stoneware casting slip	Mature Cone 6 stoneware slip with published shrinkage (X/Y/Z), water absorption % , and firing schedule . Larger pour volumes than typical small-vessel work — slip needs to flow predictably into a 12+ in master cavity without cracking on drain.	5 gal (Medium) — 15 gal (S/M/L/XL set)
2	#1 pottery plaster (large mold scale)	USG #1 Pottery Plaster (or equivalent) suitable for a 16 × 14 × 4 in two-piece pour. Need water:plaster ratio , set time , and dimensional stability (creep) over months of use.	50 lb (single Medium mold) — 200 lb (full S/M/L/XL set)
3	Mold release / mold soap	Plaster-compatible parting agent. Larger surface area than small vessels — needs to release cleanly off a 12+ in body without locking.	2 bottles / 32 oz
4			

	Cone 6 exterior glaze	Satin or matte glaze. **Color: TBD** — earth tones / iron oxide / rust red are visually closer to traditional fired Igbo-clay ware. Glaze must NOT bridge mouth or side-hole edges (acoustic-critical).	1-2 pints per body
5	Brushable wax resist	Standard ceramic wax resist for masking mouth rim, side-hole rim, and chamber interior pre-glaze.	1 jar / 16 oz
6	Hole cutter set	Round cutters, **1.5 in to 3 in diameter** , for greenware side-hole cutting. Brass tube cutters or commercial hole-cutter discs both acceptable.	1 set
7	Diamond needle file / burr set (post-bisque tuning)	Assorted profiles, **3-10 mm** ; resin-bonded acceptable. Used to enlarge ports for pitch correction.	1 set
8	Bisque + glaze firing service (if local)	Cone 06 bisque + Cone 6 glaze fire; medium-large kiln (10 ft ³ +) can fit XL family.	2 fires per body
9	(Optional) Master printing / outsource	Large-format PLA or resin print, up to **~14 in master width** with **0.010 in dimensional accuracy** . We currently print in-house but want a backup option.	1-4 master pairs (S/M/L/XL)
10	(Optional) Rubber feet — low-profile	Self-adhesive low-profile rubber feet, ~0.5 in dia, for floor stability. Test-fit only — feet damp shell tap tone, so this is an acoustic trade-off.	3 per body
11	(Optional) Piezo / contact pickup	Contact pickup that can survive being epoxied to interior wall OR magnetically/clip-mounted externally.	1 per electric variant

What we need in your quote

- ****Unit price**** and any volume breaks
- ****Minimum order quantity****
- ****Current stock status**** and ****lead time****
- ****Shipping estimate**** to 94566 (Pleasanton, CA) — or local pickup option
- ****Material safety data sheet (MSDS)**** and ****technical data sheet**** for items 1, 2, 4, 5
- ****Recommended substitutions**** if anything goes out of stock
- For item 1: most recent ****shrinkage test data**** for your batch (we re-measure on our side, but a baseline matters)
- For item 9: ****photos of similar-sized prints**** showing layer-line consistency and dimensional check process

Acceptance criteria

The prototypes are measured for X/Y/Z shrinkage, fired chamber volume (water-fill), wall-thickness consistency at rim/shoulder/belly/foot, and acoustic

tuning at TWO conditions: mouth-tone (mouth open, side covered) and side-hole tone (side open, mouth covered or hand-muted). Target tuning tolerance: ****±50 cents post-bisque**** on both modes for the Medium body. Slip with unknown or wildly variable shrinkage invalidates the master scale factor (currently $\frac{1}{1-0.12} = 1.136$), so reproducibility data matters more than headline price.

The repo is a private R&D portfolio repo; we do not need any IP-restricted documents. Public-domain MSDS and supplier spec sheets are sufficient.

Cultural note for sales / marketing teams

The udu has Igbo (Nigerian) roots and is not a generic ceramic bass pot. We're not asking for marketing positioning — we're asking for materials. But please don't include "decorative African pot" or similar reductive copy in any spec sheet response. Lineage attribution belongs to the player and the documentation, not the supplier.

Thank you,

Tony Koop

Mechanical R&D Engineer

tonykoop@gmail.com

github.com/tonykoop/udu (private; access on request)

visual-bom-brief.md

Art direction for an image-forward visual BOM.

Udu Visual BOM Brief

Goal

Create a one-page visual BOM for the slip-cast udu family that shows the body, mold, ports, finishing choices, and optional electric/water variants.

Layout

- Header: "Slip-Cast Ceramic Udu Drum Family".
- Hero image: medium udu render or finished prototype.
- Family strip: small, medium, large, XL silhouettes with target pitch ranges.
- Exploded/process view: 3D printed master halves, plaster mold halves, cast body, side-hole cut, fired body.
- BOM table: item number, material/tool, quantity, spec, cost estimate, make/buy, image.
- Physics inset: top-port and side-port Helmholtz formulas.
- Measurement inset: body volume, mouth tone, side tone, wall thickness, mass.

Image Requirements

Use real photos as soon as available:

- Printed master.
- Plaster mold halves.
- Leather-hard cast before side-hole cutting.
- Side-hole close-up.
- Fired body with hand scale.
- Optional pickup or rubber-foot detail.

Generated or CAD images should be marked as placeholders until replaced by shop photos.

wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

```
(* Udu drum family Helmholtz notebook starter *)

ClearAll["Global`*"];

c = 13510; (* in/s at about 68 F *)

areaCircle[d_] := Pi*(d/2)^2;
leff[wall_, area_] := wall + 0.6*Sqrt[area/Pi];
volumeOvoid[diam_, height_, shapeFactor_] := Pi/
6*diam^2*height*shapeFactor;
helmholtzHz[area_, volume_, neckLength_] :=
  (c/(2*Pi))*Sqrt[area/(volume*neckLength)];
centsError[measured_, target_] := 1200*Log[2,
measured/target];

uduModel[name_, diam_, height_, wall_, mouthDiam_,
sideDiam_, shapeFactor_] :=
Module[{v, aMouth, aSide, lMouth, lSide},
  v = volumeOvoid[diam, height, shapeFactor];
  aMouth = areaCircle[mouthDiam];
  aSide = areaCircle[sideDiam];
  lMouth = leff[wall, aMouth];
  lSide = leff[wall, aSide];
  <|
    "Name" -> name,
    "VolumeIn3" -> v,
    "MouthHz" -> helmholtzHz[aMouth, v, lMouth],
    "SideHz" -> helmholtzHz[aSide, v, lSide],
    "IntervalSemitones" -> 12*Log[2,
helmholtzHz[aMouth, v, lMouth]/helmholtzHz[aSide, v,
lSide]]
  |>
];

models = {
  uduModel["Small", 8, 10, 0.3, 3, 2, 0.7],
  uduModel["Medium", 10, 12, 0.3, 3, 2, 0.7],
  uduModel["Large", 12, 14, 0.3, 3, 2, 0.7],
```

```
uduModel["XL", 14, 16, 0.3, 3, 2, 0.7]
};

Dataset[models]

(* First coupled-mode placeholder: replace kCouple
with measured fit. *)
coupledFrequencies[f1_, f2_, kCouple_] :=
  Sqrt[Eigenvalues[{{f1^2, kCouple}, {kCouple,
f2^2}}]]];
```

README.md

Project artifact.

Udu Drum Family

Engineering documentation and parametric design table for the udu — the Nigerian Igbo clay vessel drum, played by alternating between the side opening and the top hole.

Part of the [tonykoop/instrument-maker](https://github.com/tonykoop/instrument-maker) catalogue.

> CAD renders, Wolfram notebook recordings, and a finalized build method are forthcoming.

License

[CC BY 4.0](LICENSE) — see LICENSE for details.